

Activity 1 — Real-World Manual Tracking vs the Pipeline



Fieldwork — track something real, by hand, for 5+ days

The real work: the automated pipeline is half of it. The OTHER half is **you, by hand**, tracking something real over several days and checking whether the automation matches reality. Machines lie silently; the analyst verifies. *Ikaw mismo ang mag-track araw-araw, tapos ihambing sa automation.*

Choose something numeric you can observe daily and **track it by hand for at least 5 separate days** (or 5 real readings over time): a stock price, fuel prices at a local station, market prices, weather readings, a currency rate. Log each with the **date** and **where** you got it.

PROOF (both required):

- **Dated PHOTOS** — one per reading/day — showing the real source at the moment you recorded it (the price board, the screen, the station sign), so the log is clearly kept over real days, not backfilled.
- A short **VIDEO (15–30 sec)** explaining your tracking: what, where, how often.

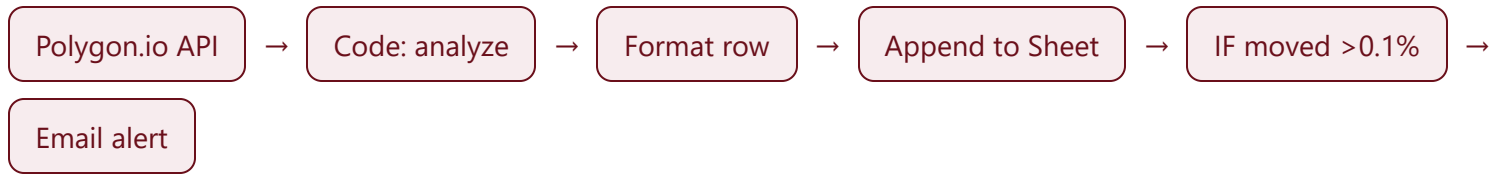
Then run the analysis notebook and **ADD a comparison:** put your hand-tracked numbers beside the automated / pipeline (or sample) numbers and note where they differ. The labor is the daily discipline and the reconciliation.

Interpretation: did your manual readings match the automated data? Where and why did they differ? Which would you trust to show a client, and why?

Shared rules: photo AND video both required (no proof = not accepted) · pairs/groups may collect together but each submits their own · reduced/desk data only with teacher's prior permission (message first) · your raw tracking log is a required deliverable.

The automation half — n8n pipeline

This guide also covers the **automation half** of Activity 1: an **n8n** workflow that fetches daily stock data from **Polygon.io**, analyzes it, logs it to a **Google Sheet**, and emails an alert. The **analysis half** (reading that sheet in Colab) is in the companion notebook.



Before you start

- A free **Polygon.io** account and API key.
- An **n8n** instance (cloud or self-hosted).
- A **Google Sheet** (and Gmail) connected to n8n.

API key safety: never share or commit your Polygon key. In the URL below, replace YOUR_POLYGON_API_KEY with your own key inside n8n only. If a key is ever exposed publicly, **rotate it** in your Polygon dashboard.

Data source — Polygon.io

Previous-day aggregate for a ticker:

```
https://api.polygon.io/v2/aggs/ticker/AAPL/prev?adjusted=true&apikey=YOUR_POLYGON_API_KEY
```

Tickers used in the demo: **AAPL** (Apple), **META** (Meta), **MSFT** (Microsoft), **GOOGL** (Google), **TSLA** (Tesla).

Step 1 — Code node: analyze the stock data

Extracts open/close/high/low/volume, computes daily return and volatility, and produces a trading signal.

N8N CODE NODE (JAVASCRIPT)


```
    signal: signal, confidence: confidence, risklevel: risklevel
  },
  recommendation: `${signal} with ${confidence} confidence. Risk level: ${risklevel}`
}
});
```

Step 2 — Code node: shape the row for the Sheet

N8N CODE NODE (JAVASCRIPT)

```
const data = $input.first().json;

return [{
  json: {
    DATE: data.date,
    SYMBOL: data.symbol,
    CLOSE_PRICE: data.prices.close,
    DAILY_RETURN: data.analysis.dailyReturn + "%",
    SIGNAL: data.analysis.signal,
    RISK_LEVEL: data.analysis.risklevel
  }
}];
```

Then add a **Google Sheets** → **Append row** node that writes these columns to your sheet.

Step 3 — IF node: only alert on real movement

```
{{ $json.analysis.dailyReturn }} is greater than 0.1
```

Step 4 — Email subject (HTML)

```
STOCK ALERT: {{ $('Append row in sheet').item.json.SYMBOL }} {{ $('Append row in sheet').item.json.DATE }}
```

Step 5 — Email body (HTML)

```
<h2>STOCK MOVEMENT ALERT</h2>

<p><strong>Symbol:</strong> {{ $('Append row in sheet').item.json.SYMBOL }}</p>
<p><strong>Current Price:</strong> ${{ $('Append row in sheet').item.json['CLOSE PRICE'] }}
</p>
<p><strong>Daily Return:</strong> {{ $('Append row in sheet').item.json['DAILY RETURN %'] }} %
</p>

<h3>SIGNAL: {{ $('Append row in sheet').item.json.SIGNAL }}</h3>
<p><strong>Confidence:</strong> {{ $('Code').item.json.analysis.confidence }}</p>
<p><strong>Risk Level:</strong> {{ $('Append row in sheet').item.json['RISK LEVEL'] }}</p>

<p><strong>Recommendation:</strong> {{ $('Code').item.json.recommendation }}</p>

<hr>
<p><em>Automated by n8n Data Pipeline</em><br>
<em>Generated at:</em> {{ $('Append row in sheet').item.json.DATE }}</p>
```

Next: let the pipeline run for a while so the sheet fills with rows, then open the **Activity 1 Colab notebook** and point it at your sheet to analyze the collected data.

Proof & Submit — required to unlock the next activity

Submit these on the activity page (the "Paste your link" box):

1. Your notebook / Colab link (the analysis + manual-vs-automated comparison).
2. Your RAW hand-tracked log (with dates + sources).
3. Your PROOF: the dated PHOTO(s) **and** the short VIDEO of your tracking.
4. Your written interpretation (did manual match automated? where/why differ?).

No fieldwork proof = the activity is NOT accepted, even if the code runs. *Kung hindi kayo makakapag-track araw-araw, mag-message muna sa teacher BEFORE the activity.*

Under the DPA / privacy notice: only collect data people agree to give; for photos/videos of people, ask their okay first; don't collect sensitive personal info you don't need.

